# Spatial lag model for spatial effects

## Parametrization

The `slm` model is defined as

$$\mathbf{x} = (I_n - \rho W)^{-1}(X\beta + \varepsilon)$$

where $I_n$ is the identity matrix of dimension $n \times n$, $W$ is a spatial weights matrix, $X$ is a matrix of covariates, $\rho$ is a spatial autocorrelation parameter, $\beta$ are coefficients of the covariates and $\varepsilon$ is zero mean Gaussian noise with precision $\tau$.

## Hyperparameters

This model has two hyperparameters $\theta = (\theta_1, \theta_2)$. The precision parameter $\tau$ is represented as

$$\theta_1 = \log \tau$$

and the prior is defined on $\theta_1$. The spatial autocorrelation parameter $\rho$ is represented as

$$\rho^* = \frac{\rho - \rho_{\min}}{\rho_{\max} - \rho_{\min}}$$

and then

$$\theta_2 = \log(\rho^*/(1 - \rho^*))$$

and the prior is defined on $\theta_2$. Here, $\rho_{\min}$ and $\rho_{\max}$ are lower and upper limits of the legal range for $\rho$.

## Specification

The slm model is specified inside the `f()` function as

```
f(<whatever>, model="slm",
       args.slm = list(rho.min = NULL,
                       rho.max = NULL,
                       X = NULL,
                       W = NULL,
                       Q.beta = NULL))
```

`args.slm` is used to define the slm-spesific parameters in the model.

**rho.min and rho.max** define the range in which $\rho$ can take values. Note that, $\rho^*$ is in the interval $(0, 1)$ and that it is re-scaled to the interval (`rho.min`, `rho.max`) when computing $I_n - \rho W$. Initial values on $\rho$ need to be re-scaled to the (0,1) interval.

**X** defines the matrix of covariates.

**W** defines the adjacency matrix.

**Q.beta** defines the precision of the vector of coefficients $\beta$ in the model.

**Hyperparameter spesification and default values**

**doc** `Spatial lag model`

**hyper**

- **theta1**
  - **hyperid** `34001`
  - **name** `log precision`
  - **short.name** `prec`
  - **initial** `4`
  - **fixed** `FALSE`
  - **prior** `loggamma`
  - **param** `1 5e-05`
  - **to.theta** `function(x) log(x)`
  - **from.theta** `function(x) exp(x)`
- **theta2**
  - **hyperid** `34002`
  - **name** `rho`
  - **short.name** `rho`
  - **initial** `0`
  - **fixed** `FALSE`
  - **prior** `normal`
  - **param** `0 10`
  - **to.theta** `function(x) log(x / (1 - x))`
  - **from.theta** `function(x) 1 / (1 + exp(-x))`

**constr** `FALSE`

**nrow.ncol** `FALSE`

**augmented** `FALSE`

**aug.factor** `1`

**aug.constr**

**n.div.by**

**n.required** `TRUE`

**set.default.values** `TRUE`

**pdf** `slm`

## Example

```
## Example using the Boston dataset from package spdep

require(INLA)
require(spdep)
data(boston)
```

```
## Index for the latent model
n <- nrow(boston.c)
boston.c$idx <- 1:n

## Define adjacency using a row-standardised matrix
lw <- nb2listw(boston.soi)
W <- as(as_dgRMatrix_listw(lw), "CsparseMatrix")

## Model definition
f1 <- log(CMEDV) ~ CRIM + ZN + INDUS + CHAS + I(NOX^2)+ I(RM^2) +  AGE +
  log(DIS) + log(RAD) + TAX + PTRATIO + B + log(LSTAT)
mmatrix <- model.matrix(f1, boston.c)

## Zero-variance for error term
zero.variance = list(prec=list(initial = 25, fixed=TRUE))

## Compute eigenvalues for SLM model, used to obtain rho.min and
## rho.max
e = eigenw(lw)
re.idx = which(abs(Im(e)) < 1e-6)
rho.max = 1/max(Re(e[re.idx]))
rho.min = 1/min(Re(e[re.idx]))
rho = mean(c(rho.min, rho.max))

## Precision matrix for beta coeffients' prior
betaprec <- .0001
Q.beta = Diagonal(n=ncol(mmatrix), betaprec)


## Priors on the hyperparameters
hyper = list(
        prec = list(
                prior = "loggamma",
                param = c(0.01, 0.01)),
        rho = list(
                initial=0,
                prior = "logitbeta",
                param = c(1,1)))

## Fit model
slmm1 <- inla( log(CMEDV) ~ -1 +
              f(idx, model="slm",
                args.slm=list(
                        rho.min = rho.min,
                        rho.max = rho.max,
                        W=W,
                        X=mmatrix,
                        Q.beta=Q.beta),
                hyper=hyper),
              data=boston.c, family="gaussian",
              control.family = list(hyper=zero.variance),
              control.compute=list(dic=TRUE, cpo=TRUE)
              )
summary(slmm1)

## Summary of the coefficients (at the end of the vector of random effects)
slmm1$summary.random$idx[n+1:ncol(mmatrix),]
```

```
## Re-scale rho to real scale
rhomarg <- inla.tmarginal(function(x){rho.min+x*(rho.max-rho.min)},
                          slmm1$marginals.hyperpar[[2]])
inla.zmarginal(rhomarg)

## Maximum likelihood estimate of model (used for comparison)
summary(m2 <- lagsarlm(f1, boston.c, lw))
```

## Notes

The estimates of $\beta$ are included at the end of the vector of random effects. See the example for details on how to extract them.